

Lab 2 — Agile Product Backlog (Epics & User Stories)

Restaurant Table Booking & Pre-Ordering App | Problem Statement #36 | Jira project key: RTB

PES University — Dept. of CSE • Built from the Lab 1 functional requirements • Name: _____ SRN: _____ Date: _____

1. Epics

The five Lab 1 functional requirements were grouped into five epics — major functional themes, each large enough to need breaking down and each owning a slice of end-user value.

Epic	Epic name	Description	Traces to	Stories	Points
Epic 1	Table Discovery & Reservation	Enable diners to see live table availability on a 2D floor plan and reserve a specific table for a chosen slot without any risk of double booking.	FR-001	4	26
Epic 2	Pre-Ordering & Menu Experience	Enable diners to browse the live menu and build a pre-order that is attached to their reservation and visible to the restaurant.	FR-002	4	16
Epic 3	Kitchen Preparation Orchestration	Time the kitchen work so that pre-ordered food is ready as the diner is seated, by computing cook-start times and releasing tickets to the KDS.	FR-003	4	18
Epic 4	Payments & Confirmation	Take payment for the pre-order or deposit through a PCI-compliant gateway and confirm the booking to the diner, storing no raw card data.	FR-004, NFR-002	4	18
Epic 5	Floor Operations & Reservation Lifecycle	Give the restaurant a truthful live floor plan: status changes synced in real time, no-show tables released, cancellations handled.	FR-005, NFR-001	4	21

2. User Stories

20 user stories, all in "As a ... / I want ... / So that ..." form. Story points use the Fibonacci scale (1, 2, 3, 5, 8, 13) and estimate complexity, volume and uncertainty together — not hours.

ID	Summary	User story	Priority	SP	Epic	Traces to	Sprint
1.1	View live floor plan	As a diner, I want to see the restaurant's 2D floor plan with every table colour-coded as Available, Reserved or Seated, so that I can see at a glance where I could sit before I commit to a booking.	High	8	Epic 1	FR-001 / UC-01	Sprint 1
1.2	Filter tables by slot and party size	As a diner, I want to filter the floor plan by date, time slot and party size, so that I only see tables that can actually seat my group at the time I want.	High	5	Epic 1	FR-001 / UC-02	Sprint 1
1.3	Select and hold a table	As a diner, I want to tap an available table and have it held for me for 10 minutes, so that nobody else takes it while I finish choosing my food and paying.	High	5	Epic 1	FR-001 / UC-01	Sprint 1
1.4	Reject overlapping reservations	As a restaurant manager, I want the system to reject any second booking for a table whose dining window overlaps an existing one, so that two parties are never sent to the same table.	Highest	8	Epic 1	FR-001 / UC-02	Sprint 1
2.1	Browse the digital menu	As a diner, I want to browse the restaurant's live menu by category with prices and prep time, so that I can decide what to pre-order before I arrive.	High	5	Epic 2	FR-002 / UC-03	Sprint 1
2.4	Show pre-order on the manager terminal	As a restaurant manager, I want to see each booking's pre-order lines against its booking ID on my terminal, so that the floor team knows what is coming before the diner walks in.	High	3	Epic 2	FR-002 / UC-03	Sprint 1

ID	Summary	User story	Priority	SP	Epic	Traces to	Sprint
2.2	Add items to the pre-order cart	As a diner, I want to add menu items to a cart with quantity and customisation notes, so that the kitchen prepares exactly what I asked for.	High	5	Epic 2	FR-002 / UC-03	Sprint 2
2.3	Edit or remove cart items	As a diner, I want to change quantities or remove items and see the total update immediately, so that I can correct my order before paying.	Medium	3	Epic 2	FR-002 / UC-03	Sprint 2
3.2	Compute cook-start times	As a diner, I want the system to work out when each of my items must start cooking, from its prep duration and my arrival time, so that my food is ready when I sit down rather than 20 minutes later.	High	8	Epic 3	FR-003 / UC-04	Sprint 2
4.1	Pay the pre-order through the gateway	As a diner, I want to pay my pre-order amount or deposit inside the app, so that my booking is confirmed and the kitchen can commit to preparing it.	Highest	8	Epic 4	FR-004 / UC-05	Sprint 2
4.2	Store payment tokens, never card data	As a restaurant owner, I want card details captured by the gateway and kept only as a token, with TLS 1.2+ everywhere, so that a breach of our database exposes no cardholder data.	Highest	5	Epic 4	NFR-002 / UC-05	Sprint 2
4.3	Receive a digital confirmation with QR code	As a diner, I want a confirmation with my booking ID, table number, itemised pre-order and a QR check-in code, so that I can check in at the door without explaining my booking.	High	3	Epic 4	FR-004 / UC-05	Sprint 2
3.1	Maintain item prep durations	As a restaurant manager, I want to set and update the standard prep duration for every menu item, so that the timeline reflects how my kitchen actually cooks.	Medium	3	Epic 3	FR-003 / UC-04	Backlog
3.3	Release the ticket to the KDS on time	As a kitchen staff member, I want the pre-order ticket to appear on the Kitchen Display System at the computed cook-start time, so that I start each dish at the right moment without watching the clock.	High	5	Epic 3	FR-003 / UC-04	Backlog
3.4	See the estimated serve time	As a diner, I want to see when my food is expected to reach the table, so that I know what to expect before I confirm the booking.	Medium	2	Epic 3	FR-003 / UC-04	Backlog
4.4	Apply a promo code	As a diner, I want to apply a promo code before the payment is authorised, so that I get the discount I was offered.	Low	2	Epic 4	FR-004 / UC-06	Backlog
5.1	Update table status on the floor terminal	As a restaurant manager, I want to move a table through Available, Reserved, Seated and Billed from my terminal, so that the floor plan always shows the true state of my restaurant.	High	5	Epic 5	FR-005 / UC-07	Backlog
5.2	Sync status changes in under 500 ms	As a restaurant manager, I want a status change on one terminal to appear on every other terminal and on the diner app in under half a second, so that two hosts never seat two parties at the same table.	Highest	8	Epic 5	NFR-001 / UC-07	Backlog
5.3	Auto-release a no-show table	As a restaurant manager, I want a reserved table to return to Available if the diner is not marked Seated within 15 minutes of the slot, so that I can sell the cover to a walk-in instead of holding an empty table.	Medium	5	Epic 5	FR-005 / UC-08	Backlog
5.4	Cancel a reservation	As a diner, I want to cancel my booking up to 60 minutes before the slot, so that I am not charged for a meal I cannot attend and the table is freed.	Medium	3	Epic 5	FR-005 / UC-08	Backlog

3. Estimation summary

Distribution	1-2 pts	3 pts	5 pts	8 pts
Number of stories	2 stories	5 stories	8 stories	5 stories

Total backlog size: 99 story points across 20 stories. Sprint 1 committed 34 and accepted 29; Sprint 2 committed and accepted 32. That leaves 38 points — the 33 not yet scheduled plus the 5-point Story 1.2 — for Sprint 3.

Why Fibonacci: the gaps widen as the numbers grow, which forces a decision between "roughly 5" and "roughly 8" instead of false precision like 6 or 7. Anything estimated at 13 or above is treated as too big and split before it enters a sprint.